

Fighting with Data: Design Implications for AI-Enabled Mission-Command Systems

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Fighting with Data argues artificial intelligence (AI)-enabled mission-command systems do not simply make existing military decision-making processes faster; they fundamentally redistribute cognition across a human-machine network. In legacy systems, commanders and staffs rely on hierarchical, sequential processes in which humans manually collect, correlate, interpret, and act on information. In AI-enabled systems, machines assume a much larger share of the cognitive labor associated with sorting data, detecting anomalies, correlating signals, and generating candidate explanations. As a result, the role of the human shifts from assembling the picture to interpreting the picture's meaning, judging risks, and deciding how to act.

This shift in the human role creates challenges in two areas of AI-enabled mission command: trust and optimization. The first challenge concerns the limits of reliabilist approaches to machine output. In conventional settings, trust can be grounded in factors such as data curation, model validation and verification, robustness, calibration, and expert oversight. But these indicators are least helpful when AI output is future oriented—when systems are used to generate plans, predictions, or assessments of enemy intent whose truth cannot be known in advance and may remain contestable even after the fact. For example, a failed plan may still be the best one available under the circumstances, whereas a successful plan may rest on spurious correlations or opaque reasoning. Regarding optimization, most current approaches focus too narrowly on the relationship between human and machine, neglecting the larger distributed cognitive system through which machine output becomes knowledge and understanding. Thus, the real design challenge is not merely to optimize model performance. Instead, the challenge is to optimize the fit of the entire human-machine system to the operational environment.

This monograph frames this problem as one of epistemic design. The central question has shifted away from whether AI can produce outputs quickly and now asks whether the entire distributed system of humans, tools, data, and processes can produce knowledge and understanding that is trustworthy enough to support decision advantage in wartime. This issue is especially important because many AI systems, particularly generative and machine-learning models, operate probabilistically and often opaquely. These systems may offer plausible recommendations without making their underlying reasoning accessible to human users. Traditional approaches to trust based on reliability, past performance, or user experience are therefore insufficient on their own.

To explain the shift toward a focus on the trustworthiness of AI systems' outputs, this monograph contrasts legacy empiricist decision making with AI-enabled reasoning. Legacy systems depend heavily on observation, induction, and deduction: Operators collect data, build confidence through corroboration, and apply rules to actions. Systems enabled by AI still use observation, induction, and deduction, but they are particularly valuable in supporting abductive reasoning—the generation and refinement of plausible explanations under conditions of uncertainty. In combat,

where commanders must often act before ambiguity is resolved, the capacity to support abductive reasoning matters more than mere speed.

In this context, mission command should be understood as a cognitively distributed system. Knowledge does not reside solely in an individual commander, staff member, or analyst; rather, it emerges from interactions among people, interfaces, models, communications networks, and the operational environment. In practical terms, fighting with data means organizing mission command so data can move across echelons and functions to generate knowledge that allows the entire system to adapt to its environment more effectively. This organization can be facilitated by three connected data layers: foundational data, which provide strategic baselines; contextual data, which support operational understanding; and situational data, which connect tactical sensing and action to the broader system.

Because AI changes the way knowledge is produced, this monograph proposes an ecocognitive framework for trust. Rather than asking only whether a system reliably produces accurate output, this framework asks whether the system fits the operational environment well enough to help commanders generate useful understanding and adapt effectively. In this view, decision advantage comes from the entire system's ability to adapt faster than an enemy, not simply to decide faster. Whether a system sufficiently fits the operational environment depends on three broad functions: representation, inference, and adaptation. Accordingly, this study proposes metrics for representational fit, inferential fit, and output and adaptability fit to evaluate whether AI-enabled systems make the battlespace more intelligible, support the formation of plausible hypotheses, and improve decision quality over time through feedback.

Success in AI-enabled mission command will depend less on computational speed than on system designs that preserve human judgment, make uncertainty visible, sustain feedback loops, and maintain cognitive resonance between human users and machine outputs. In short, the future of war fighting lies not in replacing commanders with machines; rather, it lies in building human-machine cognitive ecologies that can learn, adapt, and fight effectively with data.

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